

Program - 07

Deploy a web application to kubernetes using Minikube

Structure of the program:

Nginx-deployment.yaml

Steps:

1. Start the minikube : `minikube start`
2. Check the status of the minikube : `minikube status`
3. It shows the following status

```
_1RV24MC019_anusha_s@anu-Vivobook-Go-E1504FA-E1504FAB:~/program7$ minikube start
🐳 minikube v1.37.0 on Ubuntu 24.04
🌟 Using the docker driver based on existing profile
👍 Starting "minikube" primary control-plane node in "minikube" cluster
🚚 Pulling base image v0.0.48 ...
👤 Updating the running docker "minikube" container ...
🔧 Preparing Kubernetes v1.34.0 on Docker 28.4.0 ...
🔍 Verifying Kubernetes components...
   ■ Using image registry.k8s.io/metrics-server/metrics-server:v0.8.0
   ■ Using image gcr.io/k8s-minikube/storage-provisioner:v5
🌞 Enabled addons: storage-provisioner, metrics-server, default-storageclass
🏡 Done! kubectl is now configured to use "minikube" cluster and "default" namespace by default
_1RV24MC019_anusha_s@anu-Vivobook-Go-E1504FA-E1504FAB:~/program7$ minikube status
minikube
type: Control Plane
host: Running
kubelet: Running
apiserver: Running
kubeconfig: Configured
```

- 4 . If some the services didnt start then run the following command

`minikube delete`

`minikube start - - driver=docker`

5. Build the deployment yaml file

```
GNU nano 7.2
apiVersion: apps/v1
kind: Deployment
metadata:
  name: nginx-deployment
spec:
  replicas: 2
  selector:
    matchLabels:
      app: nginx
  template:
    metadata:
      labels:
        app: nginx
    spec:
      containers:
        - name: nginx
          image: nginx:latest
          ports:
            - containerPort: 80
```

6. Apply the nginx-deployment.yaml file

```
Kubectl apply -f nginx-deployment.yaml
```

7. Check the status

```
Kubectl get pods
```

```
Kubectl get deployments
```

8. Expose the deployment as service

```
Kubectl expose deployment nginx-deployment - --type=NodePort
--port=80
```

```
_1RV24MC019_anusha_s@anu-Vivobook-Go-E1504FA-E1504FAB:~/program7$ kubectl apply -f nginx-deploy.yaml
deployment.apps/nginx-deployment unchanged
_1RV24MC019_anusha_s@anu-Vivobook-Go-E1504FA-E1504FAB:~/program7$ kubectl get pods
NAME                                READY   STATUS             RESTARTS   AGE
mycron-29424430-btd7k              0/1     Completed          0           7h23m
mycron-29424435-c2pkj              0/1     Completed          0           7h18m
mycron-29424440-k67cj              0/1     Completed          0           7h13m
mycron-29424865-t555g              0/1     ImagePullBackOff   0           3m49s
mycron-29424870-w87w9              0/1     ImagePullBackOff   0           3m34s
nginx-deployment-6f9664446b-klpxh  0/1     ImagePullBackOff   0           7h38m
nginx-deployment-6f9664446b-pv7kl  0/1     ImagePullBackOff   0           7h38m
prog8                               1/1     Running            1 (4m12s ago) 7h20m
_1RV24MC019_anusha_s@anu-Vivobook-Go-E1504FA-E1504FAB:~/program7$ kubectl get deployments
NAME                READY   UP-TO-DATE   AVAILABLE   AGE
nginx-deployment    0/2     2             0           7h38m
```

9.Check the services

Kubectl get services

10.Access the web application

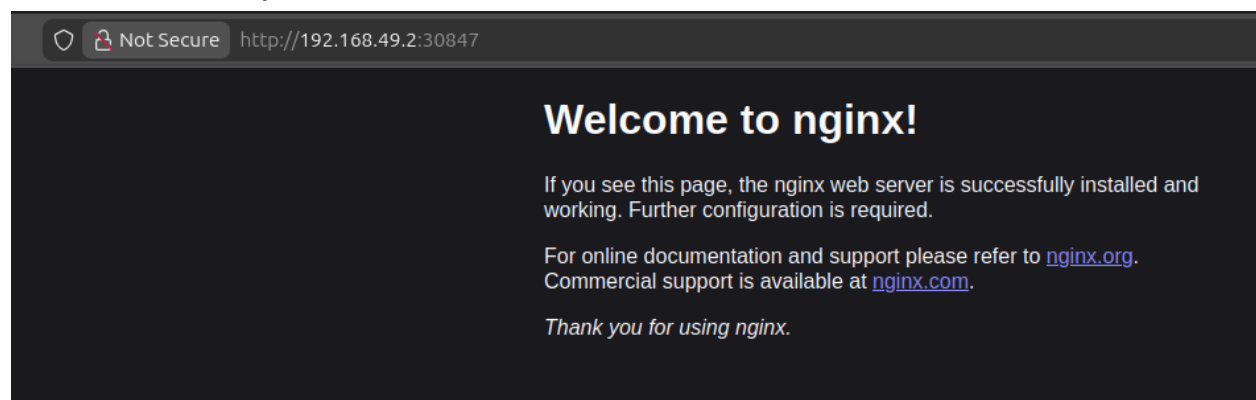
Minikube service nginx-deployment

```
_1RV24MC019_anusha_s@anu-Vivobook-Go-E1504FA-E1504FAB:~/program7$ kubectl expose deployment nginx-deployment --type=NodePort --port=80
Error from server (AlreadyExists): services "nginx-deployment" already exists
_1RV24MC019_anusha_s@anu-Vivobook-Go-E1504FA-E1504FAB:~/program7$ kubectl get services
NAME                TYPE        CLUSTER-IP   EXTERNAL-IP   PORT(S)          AGE
kubernetes          ClusterIP   10.96.0.1    <none>        443/TCP          13h
nginx-deployment    NodePort    10.106.126.75 <none>        80:30847/TCP     7h37m
_1RV24MC019_anusha_s@anu-Vivobook-Go-E1504FA-E1504FAB:~/program7$ minikube service nginx-deployment
```

NAMESPACE	NAME	TARGET PORT	URL
default	nginx-deployment	80	http://192.168.49.2:30847

🚀 Opening service default/nginx-deployment in default browser...

11.View the output in the web browser



12.Delete all the pods

Kubectl delete pods - -all

```
_1RV24MC019_anusha_s@anu-Vivobook-Go-E1504FA-E1504FAB:~/program7$ kubectl delete pods --all
pod "mycron-29425250-8fwfg" deleted from default namespace
pod "mycron-29425255-zqm5p" deleted from default namespace
pod "mycron-29425260-q5jqv" deleted from default namespace
pod "nginx-deployment-6f9664446b-klpxh" deleted from default namespace
pod "nginx-deployment-6f9664446b-pv7kl" deleted from default namespace
pod "prog8" deleted from default namespace
```

13.Delete all the services

Kubectl delete svc - -all

14.Delete all the deployments

Kubectl delete deployments - -all

```
_1RV24MC019_anusha_s@anu-Vivobook-Go-E1504FA-E1504FAB:~/program7$ kubectl delete deployment --all
deployment.apps "nginx-deployment" deleted from default namespace
```